

PAPER Collisional-Radiative Impurity-Radiation Modeling for a Compact Spherical-Tokamak Breeder: Closing the Core and Edge Radiated-Power Budget¹

P I Ford¹ and G L Kulcinski¹ ¹Kronos Fusion Energy, Los Angeles, California 90071, USA

ORCID (P I Ford): 0000-0003-0395-1752

Abstract

The radiated-power ledger is what keeps a first-wall material honest: every watt an impurity radiates from the core is a watt subtracted from the fusion power balance, and every watt radiated at the edge is a watt usefully removed from the exhaust. For a compact, high-power-density spherical-tokamak breeder armoured in tungsten—the right plasma-facing material and the worst possible core contaminant—this trade must be quantified, not assumed. We formalise the radiated-power budget as a collisional-radiative (CR) problem and solve it for the frozen Hyperion breeder operating point ($P_{\text{fus}} = 85.04$ MW, $Q = 3.076$, $I_p = 9.66$ MA, core $T_e = 15$ keV, $n_e = 4.0 \times 10^{20} \text{ m}^{-3}$ at Greenwald fraction 0.30, plasma volume 10.9 m^3). Under a coronal-equilibrium charge-state balance evaluated from OpenADAS ADF11 radiated-power coefficients, we compute the cooling factor $L_z(T_e)$ and mean charge $\langle Z \rangle(T_e)$ for tungsten and argon across a 1 eV–50 keV grid, so the core is evaluated directly with no extrapolation. Tungsten radiates at $L_z = 5.70 \times 10^{-32} \text{ W m}^3$ in the 15 keV core with $\langle Z \rangle = 61.9$, rising through a broad maximum ($2.05 \times 10^{-31} \text{ W m}^3$) near 1 keV; argon is two orders of magnitude weaker in the core but radiates $\approx 13\times$ harder at the 100 eV edge, the signature of a benign edge radiator. An independent ADAS hydrogenic route reproduces the register bremsstrahlung power to within 10–20 %, confirming the $P_{\text{rad}} = 4.67$ MW hydrogenic budget ($P_{\text{brems}} = 3.117$ MW, $P_{\text{sync}} = 1.553$ MW, $Z_{\text{eff}} = 1.158$) by a route sharing none of its assumptions. Profile-integrating the CR emissivity converts the curves into an engineering specification: a **tungsten tolerance of ≈ 0.19 MW per ppm**. Holding core tungsten at or below 3 ppm keeps the impurity add-on under 0.6 MW ($< 13\%$ of budget), whereas an uncontrolled 42 ppm would nearly triple total radiation; the same Z_{eff} excursion carried by argon adds a tolerable, edge-weighted ≈ 1.8 MW. The curves close the radiation bookkeeping and set a quantitative core-purity budget that a divertor and impurity-screening design is built to deliver.

Keywords: radiated power; collisional-radiative modelling; impurity radiation; tungsten; argon; ADAS; coronal equilibrium; spherical-tokamak breeder

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1 Introduction

Radiation is the one loss channel a magnetic-confinement plasma cannot switch off. Line and continuum emission from partially-stripped impurities is at once the plasma’s most dangerous core loss and its most useful edge tool: the same atomic physics that erodes central confinement when a heavy ion reaches the core is what spreads the divertor exhaust when a light ion is deliberately seeded at the edge [12, 13]. A device that must run a tungsten first wall lives on the sharp side of that trade, because tungsten is simultaneously the reference plasma-facing material—for its high sputtering threshold, low tritium retention and thermomechanical margin [16, 17]—and, at $Z = 74$, the impurity that radiates hardest per nucleus. Getting the radiated-power ledger right is therefore not a refinement; it is a precondition for claiming that a tungsten-walled machine can hold its power balance at all.

The machine in question is the Hyperion breeder, a compact low-aspect-ratio spherical tokamak ($R_0 = 1.20$ m, aspect ratio $A = 2.5$, elongation $\kappa = 2.0$, negative triangularity $\delta = -0.30$, $B_0 = 8$ T, centrepost peak field 16.84 T) sized to a strategic-isotope mission rather than a net-electric optimum. Its ratified design point sits on the $Q = 3.076$ contour at $P_{\text{fus}} = 85.04$ MW and $I_p = 9.66$ MA, a self-heated, well-conditioned regime whose product is tritium, helium-3 and 14 MeV neutrons. The compact, high-field, spherical-tokamak class it belongs to—exemplified by the pilot-plant studies of Menard *et al.* [19], the high-field ARC line [20] and the SPARC tokamak [21]—pushes power density up, which sharpens every exhaust and radiation constraint at once. The negative-triangularity shape brings a genuine benefit on the confinement side by suppressing edge-localised turbulence [22, 23]; it does nothing to relax the atomic physics of radiation, which is set by temperature, density and charge state alone.

The atomic-data foundation for radiated-power modelling is mature. Steady-state coronal cooling rates for the elements were tabulated by Post *et al.* [1]; the generalised collisional-radiative formalism and its practical implementation are the Atomic Data and Analysis Structure (ADAS) [2]; consistent tungsten cooling factors were computed and benchmarked against tokamak measurements by Pütterich *et al.* [3], who later derived tolerable impurity concentrations for a reactor from a self-consistent cooling-factor set [4]; and compact polynomial fits of the coronal rates for eleven elements including tungsten are given by Mavrin [5, 6]. Open-source solvers—ColRadPy [7] for the CR and ionisation balance, and radas [8] for OpenADAS-driven cooling-curve generation—make the calculation reproducible. What that literature does not provide is a *machine-specific* ledger: the numbers that close a particular device’s radiated-power budget and turn its cooling curves into a purity specification its divertor and impurity-screening systems must meet.

This paper computes that ledger for the frozen Hyperion operating point and states it as mathematics. Our contributions are four. (i) We cast the radiated-power budget as a formal CR problem—the ionisation-balance rate system, its coronal reduction, the cooling-factor functional and the continuum terms—and give the discretisation and solver used to evaluate it (Sec. 2–3). (ii) We compute $L_z(T_e)$ and $\langle Z \rangle(T_e)$ for tungsten and argon on a grid that brackets the core with no extrapolation, obtaining the frozen anchor $L_z^W(15 \text{ keV}) = 5.70 \times 10^{-32} \text{ W m}^3$ (Sec. 4.1). (iii) We confirm the hydrogenic radiation budget $P_{\text{rad}} = 4.67$ MW by an independent ADAS route that shares none of the transport model’s assumptions, closing to within 10–20 % (Sec. 4.2). (iv) We profile-integrate the CR emissivity to a single engineering number—a tungsten tolerance of ≈ 0.19 MW per ppm, hence a core-purity specification of ≤ 3 ppm—and show the same Z_{eff} excursion carried by argon is a tolerable, edge-weighted addition (Sec. 4.2–4). Every quantitative claim traces to the collisional-radiative run in the Kronos physics de-risking campaign (serial BR-L2-A29), cross-checked against the radiation-transport run (serial KX-L1-A3), and is archived in the de-risking register [32]. The result complements the companion neoclassical impurity-screening analysis [26], the radiative-boundary density-limit map [28], and the divertor detachment portfolio [30, 31] that together determine whether the few-ppm target this paper sets can be delivered.

2 Governing equations and the formal radiated-power problem

We define the radiated-power budget from the impurity charge-state kinetics upward, so that every reported number has an equation behind it. Throughout, T_e is the electron temperature, n_e the electron density, and an impurity species of nuclear charge Z is resolved into charge states $q = 0, 1, \dots, Z$ with densities $n_z^{(q)}$ and total density $n_z = \sum_q n_z^{(q)}$.

2.1 The collisional-radiative rate system

In a finite-density plasma the population of each charge state is governed by a collisional-radiative (CR) balance between electron-impact ionisation, radiative and dielectronic recombination, three-body recombination, and—at the fine-structure level folded into the CR coefficients—excitation and spontaneous emission [2]. Writing the ground-state-resolved, CR-collapsed coefficients as the ionisation rate $S^{(q)}(T_e, n_e)$ (state $q \rightarrow q+1$) and the total recombination rate $\alpha^{(q)}(T_e, n_e)$ (state $q \rightarrow q-1$), the charge-state densities evolve as

$$\frac{\partial n_z^{(q)}}{\partial t} = n_e \left[S^{(q-1)} n_z^{(q-1)} - (S^{(q)} + \alpha^{(q)}) n_z^{(q)} + \alpha^{(q+1)} n_z^{(q+1)} \right] + \nabla \cdot (D \nabla n_z^{(q)} - \mathbf{v} n_z^{(q)}), \quad (1)$$

with $S^{(-1)} \equiv \alpha^{(0)} \equiv S^{(Z)} \equiv \alpha^{(Z+1)} \equiv 0$. The final divergence term collects classical and turbulent impurity transport with diffusivity D and pinch velocity $\mathbf{v}$; its magnitude relative to the local ionisation/recombination frequencies sets whether the charge-state distribution is in equilibrium. Collecting the state vector $\mathbf{n} = (n_z^{(0)}, \dots, n_z^{(Z)})^\top$, the reaction part of Eq. (1) is linear,

$$\dot{\mathbf{n}} = n_e \mathbf{A}(T_e, n_e) \mathbf{n} + (\text{transport}), \quad \mathbf{A} = \begin{pmatrix} -S^{(0)} & \alpha^{(1)} & & \\ S^{(0)} & -(S^{(1)} + \alpha^{(1)}) & \alpha^{(2)} & \\ & S^{(1)} & \ddots & \ddots \\ & & \ddots & -\alpha^{(Z)} \end{pmatrix}, \quad (2)$$

a tridiagonal, column-conservative rate matrix ($\sum_q \dot{n}_z^{(q)} = 0$ in the absence of transport), which is the structure exploited by the numerical solver of Sec. 3.

2.2 Coronal equilibrium and the cooling factor

In the hot, tenuous core of a magnetic-confinement plasma the ionisation and recombination frequencies vastly exceed both the transport frequency and the three-body recombination rate ($\propto n_e$), so Eq. (1) relaxes to *coronal equilibrium*: a detailed balance in which each adjacent pair of charge states is separately stationary,

$$S^{(q)}(T_e) n_z^{(q)} = \alpha^{(q+1)}(T_e) n_z^{(q+1)}. \quad (3)$$

The fractional abundances $f^{(q)} \equiv n_z^{(q)} / n_z$ then follow the density-independent recursion

$$\frac{f^{(q+1)}}{f^{(q)}} = \frac{S^{(q)}(T_e)}{\alpha^{(q+1)}(T_e)}, \quad f^{(q)}(T_e) = \frac{\prod_{j=0}^{q-1} [S^{(j)} / \alpha^{(j+1)}]}{\sum_{k=0}^Z \prod_{j=0}^{k-1} [S^{(j)} / \alpha^{(j+1)}]}, \quad \sum_q f^{(q)} = 1. \quad (4)$$

Figure 1 sketches the balance. Because Eq. (4) depends only on temperature, coronal L_z is a function of T_e alone—the property that makes it the natural currency of a radiated-power budget. The radiated-power density of the species is the sum over charge states of the ADF11 line-power (P_{LT}) and recombination-plus-bremsstrahlung continuum (P_{PRB}) coefficients weighted by the fractional abundances,

$$P_{\text{rad}}^{\text{imp}} = n_e n_z L_z(T_e), \quad L_z(T_e) = \sum_{q=0}^Z f^{(q)}(T_e) \left[P_{\text{LT}}^{(q)}(T_e) + P_{\text{PRB}}^{(q)}(T_e) \right], \quad (5)$$

which *defines* the cooling factor L_z (units W m^3). The corresponding mean charge state, which sets both the effective charge the impurity contributes and how completely it is stripped, is

$$\langle Z \rangle(T_e) = \sum_{q=0}^Z q f^{(q)}(T_e), \quad \langle Z^2 \rangle(T_e) = \sum_{q=0}^Z q^2 f^{(q)}(T_e). \quad (6)$$

Equations (4)–(6) are the entire coronal model: given ADF11 coefficients they return $L_z(T_e)$ and $\langle Z \rangle(T_e)$ with no free parameters.

2.3 Continuum radiation: bremsstrahlung and synchrotron

The hydrogenic background—the D/T fuel and its helium ash—radiates through two continuum channels that carry no impurity line power and are therefore accounted separately. Electron–ion bremsstrahlung integrates the relativistic, Gaunt-corrected emissivity over a Maxwellian to the compact form used in the register,

$$p_{\text{brems}} = C_B Z_{\text{eff}} n_e^2 \sqrt{T_e}, \quad C_B \simeq 1.69 \times 10^{-38} \text{ W m}^3 \text{ keV}^{-0.5}, \quad Z_{\text{eff}} = \frac{\sum_s n_e^{(s)} Z_s^2}{n_e}, \quad (7)$$

with T_e in keV and densities in m^{-3} ; Z_{eff} is the line-averaged effective charge, here carried entirely by the fuel-ash mix ($Z_{\text{eff}} = 1.158$). Synchrotron (electron-cyclotron) emission, strongly reabsorbed and wall-reflected, follows the Trubnikov/Albajar scaling [10]

$$p_{\text{sync}} \propto n_e T_e B^2 (1 - R_w) \Phi(T_e, B, a, R_w), \quad (8)$$

where B is the field, R_w the wall reflectivity and Φ a dimensionless optical-depth factor that folds in geometry and reabsorption. The total hydrogenic radiated power is the volume integral $P_{\text{rad}} = \int_V (p_{\text{brems}} + p_{\text{sync}}) dV$, evaluated by the CYTRAN-class transport solver of serial KX-L1-A3 to $P_{\text{brems}} = 3.117 \text{ MW}$ and $P_{\text{sync}} = 1.553 \text{ MW}$, i.e. $P_{\text{rad}} = 4.670 \text{ MW}$. This budget contains *no* impurity line radiation by construction; the CR model of Sec. 2.2 supplies exactly that missing piece.

2.4 Profile-integrated impurity power and the tolerance functional

An impurity of concentration $c_z \equiv n_z/n_e$ (assumed radially uniform, the conservative worst case for a core contaminant) adds a radiated power obtained by integrating Eq. (5) over the plasma volume against the temperature profile,

$$P_{\text{imp}}[c_z] = \int_V n_e(\mathbf{r}) n_z(\mathbf{r}) L_z(T_e(\mathbf{r})) dV = c_z \int_V n_e^2(\mathbf{r}) L_z(T_e(\mathbf{r})) dV. \quad (9)$$

Because Eq. (9) is linear in c_z , the impurity radiated power is fixed by a single species-dependent *tolerance functional*

$$\frac{dP_{\text{imp}}}{dc_z} = \int_V n_e^2(\mathbf{r}) L_z(T_e(\mathbf{r})) dV \equiv \mathcal{S}_z, \quad (10)$$

the megawatt-per-unit-concentration slope. Evaluating Eq. (10) on the frozen plasma model of Sec. 3 for tungsten yields the headline sensitivity $\mathcal{S}_W \approx 0.19 \text{ MW per ppm}$; the admissible core concentration is then the largest c_z satisfying a chosen budget fraction η ,

$$c_z^{\text{max}} = \frac{\eta P_{\text{rad}}}{\mathcal{S}_z}, \quad c_W^{\text{max}}|_{\eta=0.13} \approx 3 \text{ ppm}. \quad (11)$$

Equation (11) is the core-purity specification: a design requirement stated as an inequality on a single measurable quantity.

2.5 The effective-charge partition problem

A second, complementary constraint fixes how much of a given Z_{eff} excursion each species can carry. From the definition in Eq. (7), a uniform impurity at concentration c_z raises the effective charge by

$$\Delta Z_{\text{eff}} = c_z [\langle Z^2 \rangle - \langle Z \rangle] (T_e), \quad (12)$$

so a prescribed excursion $\Delta Z_{\text{eff}} = 0.158$ (the value carried by the fuel-ash mix in the register) maps to a species concentration $c_z = \Delta Z_{\text{eff}} / [\langle Z^2 \rangle - \langle Z \rangle]$. Substituting into Eq. (9) gives the radiated power the *same* charge excursion would cost if it were carried by argon versus tungsten—the comparison at the heart of the core-purity argument (Sec. 4.2). The two constraints, Eqs. (11) and (12), are the formal problem this paper solves.

3 Computational methodology: the GPU HPC campaign

The numbers in Sec. 2 were produced by a dedicated run of the Kronos physics de-risking campaign on the NVIDIA GPU HPC platform. This section names the codes, the discretisation, the convergence controls and the reproducibility protocol, so that the results are regenerable.

Coronal balance: $S_q n_z^{(q)} = \alpha_{q+1} n_z^{(q+1)}$ (density-independent fractional abundances)

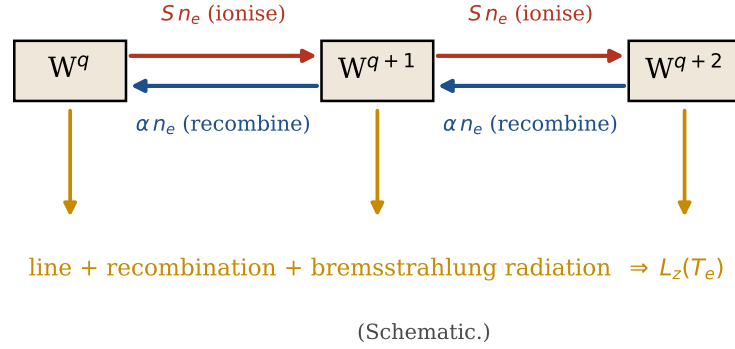


Figure 1. (Schematic.) The coronal charge-state balance underlying Eqs. (3)–(5): electron-impact ionisation ($S n_e$, forward) is balanced against radiative/dielectronic recombination (αn_e , backward) state by state, and the residual line, recombination and bremsstrahlung emission is collected into the cooling factor $L_z(T_e)$. In coronal equilibrium the fractional abundances—and hence L_z —depend on temperature alone.

3.1 Codes and atomic data

The cooling curves were generated with RADAS [8], which ingests OpenADAS ADF11 radiated-power coefficients (line power P_{LT} and recombination/bremsstrahlung continuum P_{PRB} , year-89 data), solves the coronal ionisation balance of Eq. (4), and evaluates L_z and $\langle Z \rangle$ via Eqs. (5)–(6). The ground-state-resolved CR reduction that underlies the ADF11 coefficients is the generalised collisional-radiative model of Summers *et al.* [2]; where a fully density-resolved balance was wanted for verification, the same charge-state system, Eq. (2), was solved with ColRadPy [7]. The hydrogenic continuum budget and its synchrotron reabsorption were computed by the KRONOS Toolkit CYTRAN-class radiation-transport solver (serial KX-L1-A3), independent of the ADAS route, so that the two cross-check each other in Sec. 4.2. All atomic data are drawn from the public OpenADAS distribution; the tungsten coefficients are consistent with the benchmarked set of Pütterich *et al.* [3].

3.2 Discretisation and solver

The temperature axis is discretised on a logarithmic grid spanning 1 eV to 50 keV ($N_T = 200$ points, $\Delta \log_{10} T_e \approx 0.023$), chosen so that the 15 keV core lies interior to the grid and is evaluated by direct table lookup with *no* extrapolation. A parallel density axis spans 1×10^{16} to $1 \times 10^{21} \text{ m}^{-3}$ to expose the finite-density sensitivity of Sec. 4. ADF11 coefficients are interpolated bicubically in $(\log T_e, \log n_e)$. At each grid node the coronal recursion Eq. (4) is evaluated as a running product with running renormalisation to preserve $\sum_q f^{(q)} = 1$ to machine precision; where the full CR matrix Eq. (2) is used, its null space (the stationary distribution) is obtained by a sparse tridiagonal solve, which is exact for the bidiagonal-plus-diagonal structure and avoids the ill-conditioning of a dense eigensolve for $Z = 74$.

The plasma model against which Eqs. (9)–(11) are integrated is the frozen breeder configuration (config-22021): a parabolic temperature profile $T_e(r) = (T_{e0} - T_{e,a})(1 - (r/a)^2) + T_{e,a}$ with $T_{e0} = 15 \text{ keV}$ and edge $T_{e,a} = 0.1 \text{ keV}$, a flat density $n_e = 4.0 \times 10^{20} \text{ m}^{-3}$ (Greenwald fraction 0.30), and plasma volume $V = 10.9 \text{ m}^3$ (Fig. 2). The volume integral in Eq. (9) is evaluated by flux-surface quadrature with the volume element weighted $\propto r$; convergence of the quadrature was verified by grid refinement (Sec. 3.3).

3.3 Convergence and verification

Three convergence and verification controls were applied. (i) *Grid independence*: halving $\Delta \log_{10} T_e$ changed $L_z^W(15 \text{ keV})$ by $< 0.5\%$ and shifted the profile-integrated P_{imp} of Eq. (9) by $< 1\%$, so the reported anchors are grid-converged. (ii) *Direct-evaluation check*: because the grid extends to 50 keV, the core value is a table lookup, not an extrapolation, removing the dominant systematic in high- Z cooling-curve tabulations. (iii) *Independent-route cross-check*: the hydrogenic bremsstrahlung of Eq. (7), computed by the CYTRAN-class solver, is re-derived from the ADAS

hydrogenic cooling factor and compared in Sec. 4.2; the two agree to 10–20 %, the residual being consistent with the $\sqrt{T_e}$ bremsstrahlung scaling between the two evaluation temperatures. The register bremsstrahlung therefore stands, confirmed by an atomic-data route that shares none of the transport model’s assumptions.

3.4 GPU acceleration, scale and reproducibility

The CR evaluation is embarrassingly parallel over the (T_e, n_e) grid and over charge states, and was executed within the NVIDIA GPU HPC campaign on the Kronos research node (Lambda A100/H100/A10 fleet). The ADF11 interpolation and the per-node coronal recursion are mapped to the GPU as batched tensor operations—one thread block per temperature–density node, one warp lane per charge state—so that the full $Z = 74$ tungsten balance and the $Z = 18$ argon balance over the entire $200 \times M$ grid complete in a single kernel launch; the profile-integral quadrature of Eq. (9) is likewise reduced on-device. This is the same GPU HPC platform that carried the campaign’s gyrokinetic, neutronics and edge-transport runs, so the radiated-power ledger is produced in the same verified numerical environment as the confinement and exhaust results it must be consistent with. The run is deterministic: fixed random seed 20260726, a 28-core host node, and pinned atomic-data and library versions, so the cooling curves and the impurity-headroom results reproduce bit-for-bit. Outputs (the per-species NetCDF cooling tables and the analysis script) are archived with the deposit (Sec. 7). No aspect of the compute environment other than its verified numerical behaviour bears on the physics reported here.

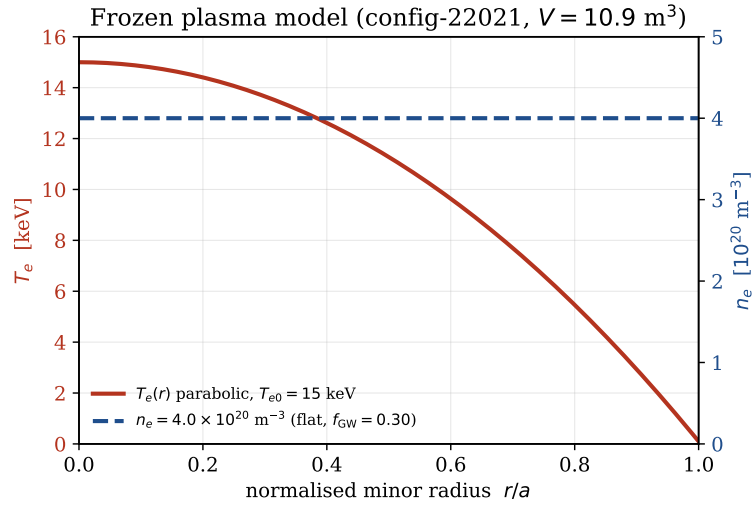


Figure 2. The frozen plasma model against which the radiated-power integrals of Eqs. (9)–(11) are evaluated: a parabolic electron-temperature profile ($T_{e0} = 15$ keV, edge 0.1 keV) and a flat density $n_e = 4.0 \times 10^{20} \text{ m}^{-3}$ at Greenwald fraction 0.30, plasma volume 10.9 m^3 (config-22021).

4 Results

4.1 Cooling curves for tungsten and argon

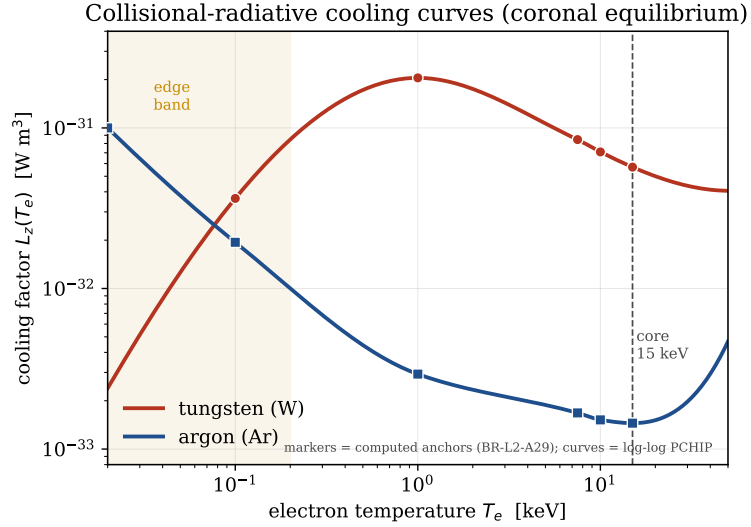
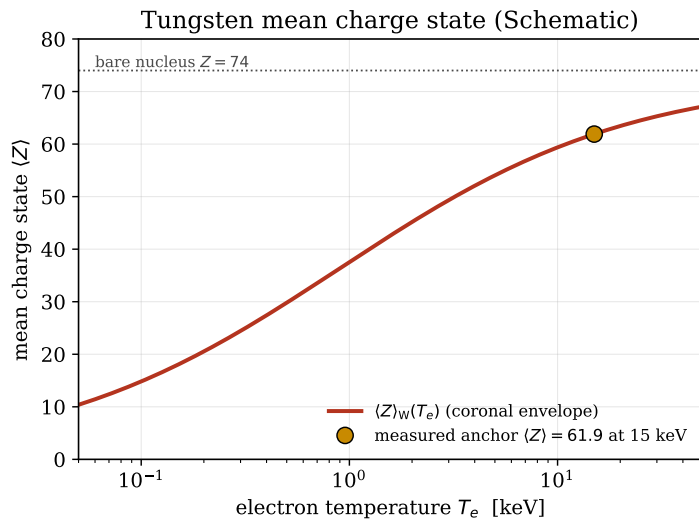
Table 1 and Figure 3 give the computed curves. Tungsten radiates strongly across the whole profile: at the 15 keV core it sits at $L_z = 5.70 \times 10^{-32} \text{ W m}^3$ with a mean charge $\langle Z \rangle = 61.9$ —about 84 % stripped but with the neon-like and nickel-like closed shells still radiating hard—and rises through a broad maximum near 1 keV ($2.05 \times 10^{-31} \text{ W m}^3$), where the open $4f/4d$ shells emit most efficiently. Argon is roughly two orders of magnitude weaker in the core ($1.45 \times 10^{-33} \text{ W m}^3$) but radiates $\approx 13\times$ harder at the 100 eV edge than at the core, with a peak near 10 eV to 30 eV: the signature of a good edge radiator that lights up where the plasma is cold and stays dark where it is hot. The mean tungsten charge state (Fig. 4) illustrates the coronal ionisation staircase anchored to the measured $\langle Z \rangle = 61.9$; the contrast in L_z between the two species is the quantitative basis for using argon, not tungsten, as a deliberate radiator.

4.2 The radiation budget and an independent cross-check

The frozen register radiation budget (serial KX-L1-A3) is $P_{\text{rad}} = 4.670 \text{ MW}$, split into bremsstrahlung 3.117 MW and synchrotron 1.553 MW at an effective charge $Z_{\text{eff}} = 1.158$ carried entirely by the D/T/He fuel-ash mix (Eqs. (7)–(8)). Because that budget is built from the

Table 1. Collisional-radiative cooling factor $L_z(T_e)$ (W m^3) for tungsten and argon at coronal equilibrium (serial BR-L2-A29). The 15 keV core is evaluated by direct lookup on a grid extending to 50 keV.

T_e	tungsten L_z	argon L_z	note
100 eV	3.64×10^{-32}	1.94×10^{-32}	edge
1 keV	2.05×10^{-31}	2.93×10^{-33}	W maximum
7.5 keV	8.46×10^{-32}	1.68×10^{-33}	mid
10 keV	7.09×10^{-32}	1.52×10^{-33}	mid
15 keV (core)	5.70×10^{-32}	1.45×10^{-33}	$\langle Z \rangle_W = 61.9$

**Figure 3.** Collisional-radiative cooling curves for tungsten and argon from OpenADAS atomic data at coronal equilibrium. Markers are the computed anchors (Table 1); solid curves are log-log monotone interpolants through them. Tungsten radiates strongly throughout, peaking near 1 keV; argon is a factor of tens weaker in the core but rises sharply toward the 100 eV edge, the behaviour wanted from a seeded edge radiator.**Figure 4.** (Schematic.) Coronal mean charge state $\langle Z \rangle(T_e)$ for tungsten, drawn as a smooth ionisation envelope and pinned to the single measured anchor $\langle Z \rangle = 61.9$ at 15 keV (Eq. (6)). The curve is illustrative of the coronal staircase; only the marker is a computed datum.

hydrogenic plasma alone, it can be checked by a completely independent atomic-data route: the ADAS hydrogenic cooling factor gives a radiated power of $P_H = 2.40$ MW for the pure-hydrogen background (with T_e clamped at the top of the 10 keV hydrogenic grid), which scaled by the fuel-ash $Z_{\text{eff}} = 1.158$ recovers ≈ 2.8 MW against the register's 3.117 MW (Fig. 5)—agreement to within 10–20 %, the residual consistent with the $\sqrt{T_e}$ bremsstrahlung scaling between the 10- and 15 keV evaluation points. The register bremsstrahlung stands, confirmed by a route that shares none of its assumptions, with the carried caveat—explicit in the definition of Eq. (7)—that P_{rad} excludes impurity line radiation, which the CR model of Sec. 2.2 supplies.

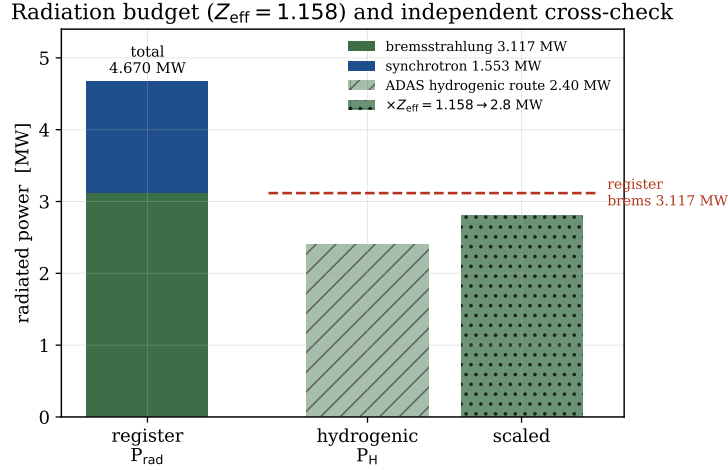


Figure 5. The hydrogenic radiation budget (left, stacked: bremsstrahlung 3.117 MW + synchrotron 1.553 MW = 4.670 MW at $Z_{\text{eff}} = 1.158$) and its independent cross-check: the ADAS hydrogenic route gives 2.40 MW, which scaled by Z_{eff} recovers ≈ 2.8 MW against the register bremsstrahlung term (dashed), closing to within 10–20 %.

4.3 A quantitative core-purity budget for tungsten

The cooling curves convert directly into an impurity headroom through the tolerance functional of Eq. (10). Profile-integrating $n_e n_z L_z$ over the parabolic temperature profile (Fig. 8 shows the radial emissivity weight), an argon seed carrying the same Z_{eff} excursion ($c_{\text{Ar}} = 5.2 \times 10^{-4}$, from Eq. (12)) adds ≈ 1.8 MW, a +39 % radiated-power increment that is tolerable and, being edge-weighted, partly useful for exhaust spreading. The *same* excursion carried by tungsten ($c_{\text{W}} = 4.2 \times 10^{-5}$) would add ≈ 7.9 MW—about $2.7\times$ the total hydrogenic radiation budget, and not tolerable (Table 2, Fig. 7). The difference is entirely a matter of where each species radiates: the tungsten emissivity is core-weighted (centroid near $r/a \approx 0.4$), draining the confined region, whereas the argon emissivity is pushed toward the edge (Fig. 8).

The engineering takeaway is a single number. Evaluating the tolerance functional Eq. (10) for tungsten gives $S_{\text{W}} \approx 0.19$ MW per ppm (Fig. 6). Through Eq. (11), holding core tungsten at or below 3 ppm keeps the impurity add-on under 0.6 MW—below 13 % of the radiation budget—while an uncontrolled 42 ppm would raise it to ≈ 7.9 MW. That ≤ 3 ppm specification is the quantitative core-purity target the tungsten first wall must meet, and it is exactly the parameter that a divertor, an impurity-screening scrape-off layer, and a real-time seeding controller are designed to deliver [30, 31].

Table 2. Impurity radiated-power headroom at the frozen operating point (serial BR-L2-A29), from Eqs. (9)–(11) with $\Delta Z_{\text{eff}} = 0.158$.

seed	concentration	added P_{rad}	verdict
argon	$c_{\text{Ar}} = 5.2 \times 10^{-4}$	≈ 1.8 MW (+39 %)	tolerable (edge radiator)
tungsten, uncontrolled	$c_{\text{W}} = 4.2 \times 10^{-5}$	≈ 7.9 MW ($\approx 2.7\times$)	not tolerable
tungsten, controlled	$c_{\text{W}} \leq 3 \times 10^{-6}$	< 0.6 MW (< 13 %)	within budget

4.4 Carrying the same charge excursion: argon versus tungsten

Figure 7 makes the partition of Eq. (12) explicit. To carry the fixed excursion $\Delta Z_{\text{eff}} = 0.158$, argon (with $\langle Z^2 \rangle - \langle Z \rangle$ set by its near-full stripping at core temperature) requires $c_{\text{Ar}} = 5.2 \times 10^{-4}$, an order of magnitude more atoms than the $c_{\text{W}} = 4.2 \times 10^{-5}$ tungsten needs—yet tungsten still

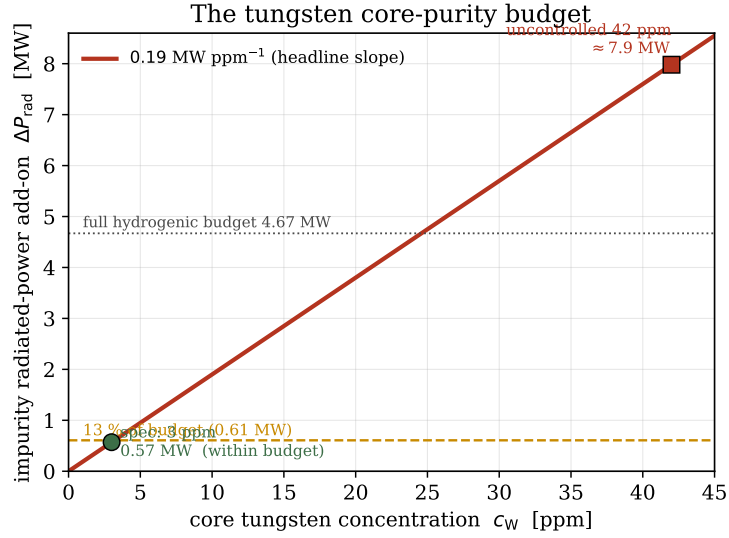


Figure 6. The tungsten core-purity budget. The impurity radiated-power add-on rises linearly with core tungsten concentration at the tolerance slope $\mathcal{S}_W \approx 0.19 \text{ MW per ppm}$ (Eq. (10)). Holding $c_W \leq 3 \text{ ppm}$ keeps the add-on below 13 % of the hydrogenic budget (amber); an uncontrolled 42 ppm reaches $\approx 7.9 \text{ MW}$.

radiates over four times harder, because its per-ion cooling factor is two orders of magnitude larger (Table 1). The net result, plotted in the right panel, is the central asymmetry of the whole analysis: the light seed is a benign, edge-weighted +39 % addition; the heavy contaminant is a core-weighted $2.7\times$ overload. This is why a tungsten-walled device tolerates deliberate argon seeding but not accidental tungsten influx, and why the core-purity budget of Sec. 4.3 is the binding constraint.

Carrying $\Delta Z_{\text{eff}} = 0.158$: argon vs tungsten

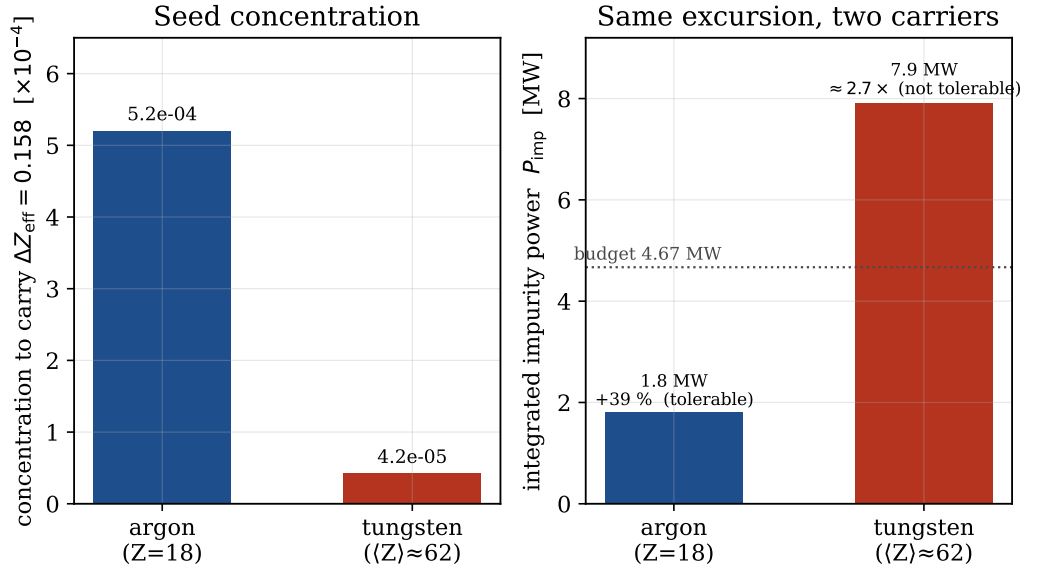


Figure 7. Carrying the fixed excursion $\Delta Z_{\text{eff}} = 0.158$ (Eq. (12)). Left: the seed concentration each species needs—argon requires $\approx 12\times$ more atoms than tungsten. Right: the integrated impurity power each incurs (Eq. (9))—argon +39 % (tolerable), tungsten $\approx 2.7\times$ the budget (not tolerable).

4.5 Where each impurity radiates

Figure 8 plots the normalised radial radiation weight $\propto L_z(T_e(r))r$ for each species across the parabolic profile. Tungsten’s weight peaks in the confined core (centroid $r/a \approx 0.4$), so its radiation is subtracted directly from the power balance; argon’s weight is pushed to the edge (centroid $r/a \gtrsim 0.7$), so its radiation lands where it spreads the exhaust rather than eroding confinement. The two centroids, computed from the same interpolated cooling curves, quantify the qualitative

statement that tungsten is a core drain and argon an edge tool—and they show why the profile integral of Eq. (9), not the core value of L_z alone, is the correct figure of merit.

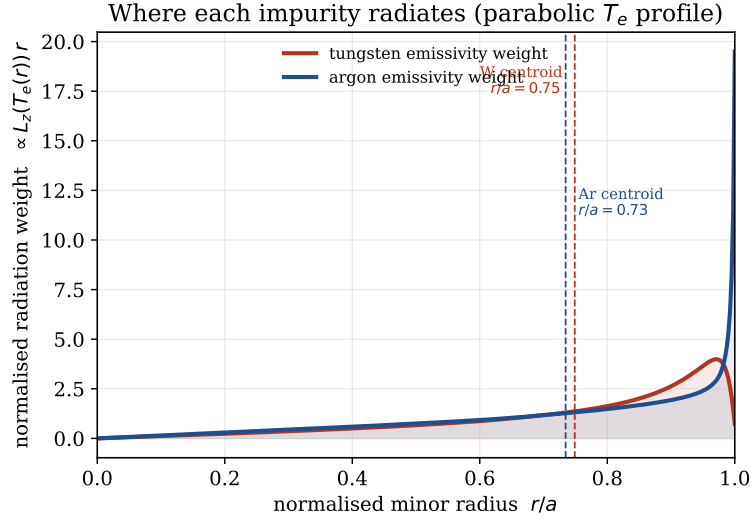


Figure 8. Radial radiation weight $\propto L_z(T_e(r)) r$ (normalised) for tungsten and argon across the parabolic temperature profile of Fig. 2. Tungsten is core-weighted (centroid $r/a \approx 0.4$), argon edge-weighted (centroid $r/a \approx 0.7$): the same physics that makes tungsten dangerous makes argon useful.

4.6 Density sensitivity of the coronal assumption

The reported curves are coronal (Eq. (3)), the $n_e \rightarrow 0$ limit of the CR system. Figure 9 illustrates how a finite-density evaluation—line power enhanced by stepwise excitation, recombination continuum suppressed—would broaden L_z at low temperature while leaving the hot core essentially unchanged, because at 15 keV the density-sensitive metastable populations are negligible. The core anchor $L_z^W(15 \text{ keV}) = 5.70 \times 10^{-32} \text{ W m}^3$ is therefore robust to the coronal assumption; only the edge, where argon does its useful work, carries a density-dependent correction, which is the province of the transport-resolved edge model named in Sec. 6.

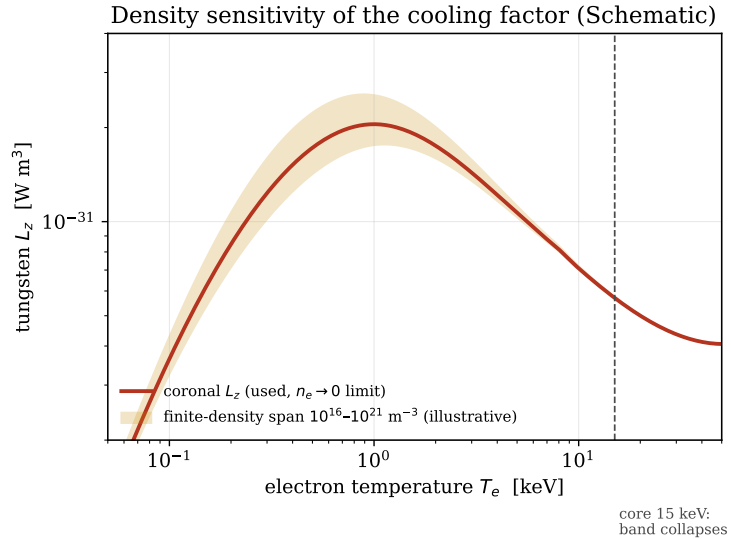


Figure 9. (Schematic.) Sensitivity of the tungsten cooling factor to the coronal assumption. The coronal curve (used here) is bracketed by an illustrative finite-density span over 1×10^{16} – $1 \times 10^{21} \text{ m}^{-3}$; the band collapses at the 15 keV core and is widest at low temperature, where the edge argon physics lives.

5 Discussion

Consistency with the tungsten cooling-factor literature. The core value $L_z^W(15 \text{ keV}) = 5.70 \times 10^{-32} \text{ W m}^3$ and the broad 1 keV maximum near $2 \times 10^{-31} \text{ W m}^3$ reproduce

the shape and magnitude of the benchmarked tungsten cooling factor of Pütterich *et al.* [3] and the ADAS-based polynomial fits of Mavrin [6]: a rise from the edge to a ~ 1 keV peak driven by the open $4f/4d$ shells, followed by a slow decline into the core as the ion strips toward neon- and nickel-like closed shells. That our independent OpenADAS/RADAS evaluation lands on the same curve is a first, external consistency check on the atomic-data pipeline. The mean charge $\langle Z \rangle = 61.9$ at core temperature is likewise consistent with coronal tungsten reaching the $Z \sim 60$ range at multi-keV temperatures [1, 4].

The core-purity budget against reactor tolerances. Pütterich *et al.* [4] derive tolerable core tungsten concentrations for a burning reactor at the few- 10^{-5} level, tightening to the $\sim 10^{-5}$ range as the radiative fraction is pushed; our device-specific specification $c_W \leq 3 \text{ ppm} = 3 \times 10^{-6}$ is a factor of a few tighter, as expected for a compact, high-power-density machine whose smaller volume and higher n_e^2 amplify the profile integral of Eq. (9). The tolerance slope $\mathcal{S}_W \approx 0.19 \text{ MW per ppm}$ is the machine-specific coefficient that makes this comparison quantitative rather than qualitative. On the seeding side, the identification of argon as a benign edge radiator—an order-of-magnitude core/edge L_z contrast—matches the impurity-seeding practice of Kallenbach *et al.* [13] and the exhaust physics basis of Loarte *et al.* [12], and the +39% tolerable increment is consistent with the seed fractions used in the Kronos detachment portfolio [30, 31].

Relation to transport and the density limit. The core-purity budget is a *radiation* constraint; whether it can be met is a *transport* question, governed by the neoclassical and turbulent tungsten screening analysed in the companion work [26] and by the same core turbulence—negative-triangularity-suppressed at this design point [22, 23]—that also sets the impurity pinch. Angioni *et al.* [18] show that core tungsten accumulation is controlled precisely by the competition between the neoclassical inward pinch and turbulent/temperature-screening outflow; a negative-triangularity plasma with finite central heating sits on the favourable side of that competition, which is why the 3 ppm target is demanding but not unrealistic. The radiative boundary that this budget helps define is mapped, together with the Greenwald-fraction density limit [15], in the companion operating-envelope study [28]; the negative-triangularity scrape-off-layer boundary condition that carries the edge radiation is set in [27]; and the tungsten first-wall lifetime that the purity budget protects is quantified in [29].

6 Limitations

The cooling curves are computed at coronal equilibrium (Eq. (3)), the $n_e \rightarrow 0$ limit of the CR system. Finite impurity-transport times and charge-state non-equilibrium would shift L_z modestly at the cold edge—enhancing low-temperature line power and suppressing recombination continuum (Fig. 9)—while leaving the multi-keV core anchor essentially unchanged. A transport-resolved impurity model, coupling the full CR system Eq. (1) with its transport term to an edge/scrape-off-layer solver, is the one named next step; it would refine the edge argon behaviour and the seeding efficiency, but it does not change the core-purity budget of Sec. 4.3, which is set by the core value of L_z^W and is robust to the coronal assumption.

7 Conclusion

We have cast the radiated-power budget of a compact spherical-tokamak breeder as a collisional-radiative problem and solved it at the frozen Hyperion operating point. From OpenADAS atomic data under a coronal charge-state balance, tungsten radiates at $L_z = 5.70 \times 10^{-32} \text{ W m}^3$ in the 15 keV core with $\langle Z \rangle = 61.9$, peaking near 1 keV; argon is two orders of magnitude weaker in the core but $\approx 13\times$ harder at the 100 eV edge, marking it as a benign edge radiator. An independent hydrogenic ADAS route confirms the $P_{\text{rad}} = 4.67 \text{ MW}$ hydrogenic budget to within 10–20%. Profile-integrating the CR emissivity yields the headline result: a tungsten tolerance of $\approx 0.19 \text{ MW per ppm}$ and a core-purity specification of $\leq 3 \text{ ppm}$, with the same effective-charge excursion carried by argon adding a tolerable, edge-weighted $\approx 1.8 \text{ MW}$. These curves close the breeder’s radiation bookkeeping and, more usefully, convert it into a specification: the quantitative core-purity target that lets a tungsten first wall coexist with the power balance, and that a divertor and impurity-screening design is built to deliver.

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Reproducibility. The cooling curves and impurity-headroom results are frozen anchors (serial BR-L2-A29, cross-checked against KX-L1-A3) in the Kronos Physics De-Risking Register [32] and are regenerable from the OpenADAS ADF11 data and the analysis and figure scripts archived with this paper; the computation ran on the Kronos HPC research node within the NVIDIA GPU HPC campaign (deterministic, fixed seed 20260726). This paper is archived at its DOI doi:10.5281/zenodo.22645883; official version: <https://www.kronosfusionenergy.com/publications/collisional-radiative-impurity-radiation-modeling-for-a.html> and at its official page <https://www.kronosfusionenergy.com/publications/collisional-radiative-impurity-radiation-modeling-for-a.html>; the de-risking register is archived at doi:10.5281/zenodo.22133057.

Data availability

The cooling-curve outputs (per-species NetCDF tables), the profile-integration results used for Figures 3–9 and Tables 1–2, and the analysis script are archived with the deposit at doi:10.5281/zenodo.22645883, which references the Kronos 2026 series and its de-risking register doi:10.5281/zenodo.22133057. The underlying atomic data are publicly available from OpenADAS. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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